

A Practical Mobile Robotics Engineering Course Using LEGO Mindstorms

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Abstract. Mobile robotics and autonomous systems are subjects of increasing interest in the curricula of engineering studies, being application domains of growing importance in leading research and industry trends. In order to fulfill this demand, we have designed a robotics course for students from different engineering degrees and backgrounds. The course includes basic ideas from main topics in mobile robotics, with an important practical component. During the course, each student team is provided with a LEGO Mindstorms kit to freely build a robot and develop several modules demonstrating the specific competences being taught. Complete integration of all modules is the last step towards participation in a final obstacle-race competition, which has been shown to be a very successful and motivating activity for the students. This paper summarizes the course content, goals and methodology and details the results obtained in the student contest performed in the recent years.

Keywords: Mobile robotics course, student contest, LEGO Mindstorms.

1 Introduction

Robotics applications have become widespread in recent years, moving from confined industrial environments to daily life situations. Increasingly automated cars, rescue vehicles for distress situations, home robots that aid in housekeeping tasks, envisioned assistants for the elderly are a few examples of present and eagerly anticipated robotic presence in our lives.

In particular, our research group has strong expertise in the fields of autonomous and service robotics, a core competence in this kind of applications. Therefore, we have designed different activities related to engineering and service robotics, to push and promote both the engineering studies and the robotics field among current and prospective students. The focus of this paper is an undergraduate course, whose main goal is to provide students with basic concepts and skills about mobile and autonomous robotics. We teach well-known algorithms used to solve basic tasks in these systems and to provide students with a robotic platform where they can practice and test all these ideas.

In educational frameworks, the typically high cost of robotic hardware can be an obstacle in the way of designing practical activities for students, that could otherwise be addressed by means of simulations. However, the use of real robots provides a richer challenge for students, that have thus to tackle the varied problems that appear in complex systems involving sensors, actuators, embedded computers and software; besides this offers them a more fulfilling and engaging experience. In recent years, some toy-like products have reduced this entry gap enormously, providing most of the characteristics that exist in standard research robots in packages more suitable and affordable for education. In particular, we have focused our activities on the LEGO Mindstorms platform. The Mindstorms robots have already been used in didactic activities at a broad range of academic levels: from school kids [2] to multi-robot research projects [1]. We have leveraged this platform as a key resource to build upon in our Service Robotics course, which is taught at the several engineering degrees described next.

1.1 Framework of Our Robotics Course: Engineering School and Related Degrees

Our engineering school at the University of Zaragoza has 2900 students enrolled in five major engineering degrees. Besides, there are 300 postgraduate and PhD students. The robotic field is of great interest for many engineering degrees; therefore, our robotics course is offered as a common course for students from the different degrees detailed next.

The engineering degrees offered in our school are currently in transition from the older structure to the new European Bologna education plans. Currently, engineering studies consist of 5 academic years, 300 ECTS, with a level of competences equivalent to the new Master level, according to the European Higher Education Area. The robotics course presented here is currently offered at the following engineering degrees at the CPS: *Industrial Engineering*, *Telecommunications Engineering* and *Computer Science Engineering*.

As mentioned above, current degrees are being adapted to the European Higher Education Area. In this new framework, our university is offering several bachelor degrees (4 years, 240 ECTS), and the robotics course is planned to be part of last years courses in many of them: *Electrical Engineering*, *Mechanical Engineering*, *Electronics and Automation Engineering*, *Industrial Technologies Engineering*, *Telecommunication Technologies and Services Engineering* and *Computer Engineering*.

2 Mobile Robotics with LEGO Mindstorms

The LEGO Mindstorms platform has already been used at a broad range of activities regarding not only robotics learning and dissemination but other engineering topics. There are from robotics research results based on this platform

[1],[5] to multiple teaching experiences in different academic levels and different subjects (control, mobile robotics or programming) [9],[7],[3],[6],[13]. Recent studies are dedicated to evaluate up to which level this kind of platforms are an effective motivation [10]. This is a fact that we can easily guess and check when we propose students to work with them, not only for specific robotics, but to learn other related subjects [8].

2.1 LEGO Mindstorms Platform

We use the LEGO Mindstorms, version NXT 2.0, as robotic platform to run the experimental part of our course. This robotic platform is composed by its core part, the NXT brick that contains an ARM microprocessor, and a set of sensors and servo-motors that can be connected to the main block. We can connect up to 3 motors and 4 sensors from a wide range of possibilities, e.g., we work with reflected light, sonar, sound, touch, gyro, compass and camera sensors. Fig. 1 shows the NXT brick with the standard components connected to it and details of some additional sensors we use in the course projects. The NXT brick contains a small display to edit, control and launch programs and to allow the running program to display information. The programming can be done both directly at this console or, more often, from a computer. Once the program is compiled in the computer, it can be transferred to the NXT through USB or Bluetooth.

There are plenty of possibilities to program the Mindstorms, but as main development platform in our practical sessions, we use RobotC [12]. This programming environment also provides access to the NXT brick file system and allows uploading and debugging programs on the brick (shown in Fig. 2). This framework uses the industry standard C-programming language and additional language extensions specifically designed for sensors and motor usage.



Fig. 1. Mindstorms standard components connected to the NXT brick and sample robot models (left) and details of some non standard used sensors (right)

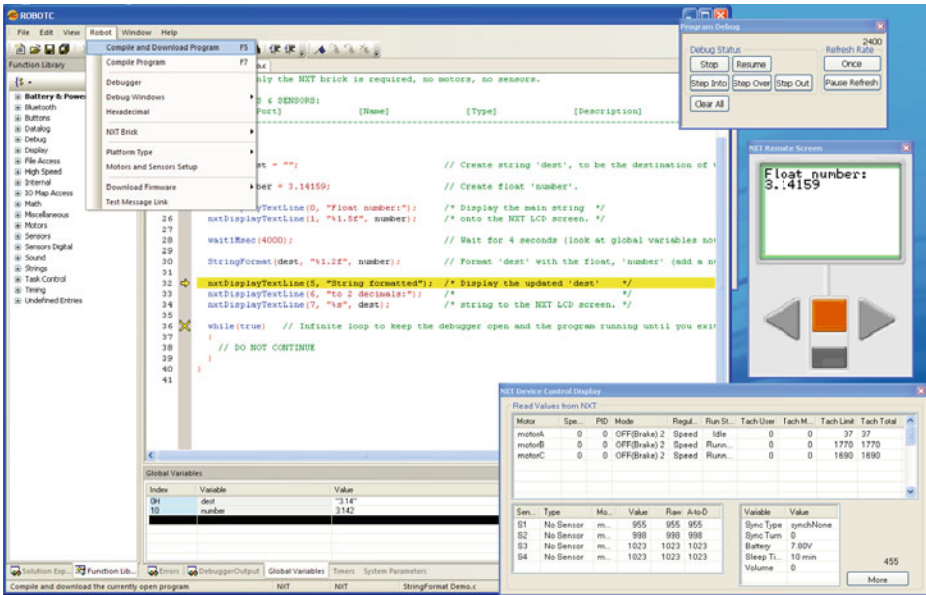


Fig. 2. Robot C programming environment

3 Our Course: Service Robotics

The course aim is to explain and develop different aspects and technologies of the so-called Service Robotics, which is an increasingly popular field with many new applications. The Service Robotics field is devoted to most robotic applications out of typical Industrial Robotics, where traditional robotic arms or manipulators are applied in automated industrial processes.

As previously mentioned, this course is offered to different engineering studies, such as Telecommunication, Computer Science, Electronic and Automation and Industrial Engineering. Therefore, the multidisciplinary orientation is inherent to this course.

The course has a strong practical orientation and the student is encouraged from the beginning to freely propose solutions and alternatives in mechanical aspects, programming techniques, sensors to be used, and so on. However, there is another important objective: the students should learn the basic theoretical foundations of robotics, in such a way that they are able to apply them. More detailed goals, methodology and course components are described in the following subsections.

3.1 Course Goals and Methodology

This section describes first the goals of the course, regarding the competences that will be developed and the results that we expect from the students after

passing the course. Secondly, the methodology and activities developed through the course are summarized.

Competencies developed and expected results. This course aims to make the student develop some general skills related to engineering and robotics and acquire some specific capabilities in robotics. The specific capabilities are directly related to the contents learned in the course, while the general competencies developed are the following:

- Capability to combine general and specific knowledge in order to propose innovative solutions to technological problems.
- Problem-solving skills, with initiative, creativity and criticism.
- Oral and written knowledge communication.
- Capability of continuous and autonomous learning.
- Team work.

The learning results of the course, i.e., what every student who passes the evaluation should achieve, are the following:

- Knowing the basics and typical applications of intelligent robotic systems.
- Understanding perception techniques in robotics and how to apply them.
- Being able to implement algorithms to model the environment and localize the robots on it.
- Being able to apply path planning and navigation techniques in simple environments.
- Being able to select the most suitable robotic framework for a particular application.
- Being able to understand a research paper on the topics included in the course and explain the content to other students.

Methodology. The methodology applied is based on three completely integrated pillars: classroom sessions, tutored laboratory sessions and group project. In the first one, the basic concepts and technologies are described in such a way that the students learn the solid scientific and technical foundations of robotics to be applied in the practical sessions and in the group project. In the second one some tutored practical sessions are developed in the laboratory using the real robot and sensors. These sessions are associated to the classroom sessions and developed after each group of conceptual blocks. The results of each practical session are used by the students in the third pillar, the group project, as pieces to build and integrate in the robots the methods learned. In the laboratory sessions and the practical work the students are organized in pairs, preferably coming from different engineering backgrounds to be complementary. Both the results of tutored sessions and the group project are evaluated. Finally, a part of the evaluation is based on a competition or contest between groups. We think this additional part of the evaluation acts as an incentive for improving the resulting autonomous robot capabilities. This way, the students feel more involved in the learning process, try to understand better the theoretical concepts transmitted, and their creativity is encouraged. So far, the experience confirms these expectations.

3.2 Course Program - Classroom sessions

The classroom sessions are devoted to introduce the basic concepts and techniques in robotics in order that the students are able to develop a complete autonomous robot. Diverse aspects are developed in this multidisciplinary course: mechanics, control, programming, perception systems, trajectory generation, specific robotic techniques, and a general vision about the state of the art of Service Robotics. The course aims to simultaneously transmit solid scientific and technological foundations and practical methods to apply them. Many of the studied techniques are complemented with exercises which are oriented to reinforce the theoretical parts and to prepare the practical sessions in the laboratory in a rigorous way. The exercises are solved and presented by the students, and discussed with the professor and the rest of the students.

An important characteristic we seek in the course is to design a program that students with different backgrounds (electric, automation, computer science, communications engineering) are able to follow. The course will provide the students with the concepts and techniques to be applied in the laboratory sessions. Besides, when designing the program, it was very important to synchronize properly the classroom sessions with the practical sessions, to make sure the theoretical basis have been previously analyzed in the classroom sessions. Following these criteria, we have developed the course program into the following blocks:

1. *Introduction.* A general perspective about the state of the art in Robotics is described. A historical evolution of robots, techniques, and applications is presented, and the most relevant results in the last years are shown.
2. *Service Robotics.* The major questions to be solved in autonomous robots are presented and justified. The closed loops perception-localization and mapping-task and motion planning-motion control-locomotion and actuation are described in a general way, providing some hints about how each of these problems can be tackled.
3. *Spatial representation, locomotion and kinematics.* The mathematical tools to manage the object and robot absolute and relative localization, transformation equations, and motion computation, both in 3D and 2D, using these tools are developed. The different types of robots from the point of view of locomotion mechanisms are described and the involved geometric and kinematic models are presented.
 - (a) Homogeneous representation.
 - (b) Translation and rotation in the space.
 - (c) Reference systems and transformations.
 - (d) Types of robots. Motion and drive mechanisms and models.
4. *Concurrent Processes* The basic notions about programming of parallel processes, mutual exclusion and semaphores are explained. The objective is that

the students are able to model and program tasks in the language used in the next practical sessions, such as continuous odometry computation or range sensors testing for collisions.

- (a) Task execution and parallel processes.
 - (b) Exclusive access and semaphores.
5. *Perception.* A description of the main internal and external sensors used in robotics to compute the motion and the localization in the environment is presented. A special focus on visual sensor is given, since one of the practical sessions is devoted to the use of a camera on the robot.
 - (a) Odometry and inertial sensors.
 - (b) Range sensors.
 - (c) Computer Vision in Robotics.
 - (d) Other sensors.
 6. *Autonomous Navigation.* Methods for trajectory generation, control of the motion, and moving object tracking are described. The basic path planning and reactive navigation techniques are discussed, maintaining the focus on the ones that can be implemented in the robots used in the laboratory sessions.
 - (a) Automatic motion generation and tracking of trajectories.
 - (b) Path planning.
 - (c) Reactive navigation.
 7. *Localization and environment modeling.* Localization, map building and basic SLAM techniques from the information of the sensors presented in previous blocks are described. The techniques are also adapted to the specific hardware used in the laboratory sessions.
 - (a) Localization.
 - (b) Map building.
 - (c) Self-localization and map building with different sensors.
 8. *Types of Mobile Robotics architectures.* The different software architectures proposed and used in mobile robotics are described and related.
 - (a) Deliberation systems.
 - (b) Reactive systems.
 - (c) Hybrid systems.
 - (d) Behaviour-based systems.

The temporal distribution of these chapters/blocks and the practical sessions, detailed in next subsection, is summarized in Table 1.

3.3 Course Program - Practical sessions

This course has a large practical component, including four guided sessions and a robotics group project summarized next. As described, we chose the Robot C framework [12] for the development of these practical sessions with the LEGO Mindstorms robots.

Table 1. Temporal distribution of classroom and laboratory sessions in the course

weeks	classroom (3h/week)	laboratory (3h/week)
1 st	Chapters 1&2	
2 nd – 3 rd	Chapter 3	P1
4 th	Chapter 4	Hours to finish designing the robots
5 th – 6 th	Chapter 5	P2
7 th	Chapter 6	P3
8 th – 9 th	Chapter 7	P4
10 th – 14 th	Chapter 8 & invited seminars	Autonomous work and tutoring hours to integrate the final group project
15 th	Student results presentations	Competition event

P1 - Initial practical session. This session consists of designing and building a mobile robot from a Mindstorms kit and getting used to the development environment and programming language that will be used in the course, Robot C. Every group will be provided with a Mindstorms kit and should design and build a free-design robot that should just fit a few requirements to be able to perform the compulsory tasks of the rest of practical sessions. Besides, a few simple exercises should be done to get to know the Robot C programming environment and the range, quality and way of accessing the robot sensors.

P2 - Motion generation and odometry estimation. This session aims to practice the advanced motion generation and estimation (from chapters 5a and 6a) and the design of concurrent processes (chapter 4). Each group has to program a module for their robot that estimates the odometry of the robot as it moves. This task must be launched in parallel with the main program of the robot. In this case, the main task of the robot will be performing a pre-defined trajectory, including some curved paths, such as an 8-like trajectory.

P3 - Vision based object tracking. This session has been designed as an introduction to vision based tasks, such as tracking, in robotics (from chapter 5c). In particular, a simple object tracking will be implemented and tested, augmenting the modules of the robot with object recognition and object tracking functions.

P4 - Autonomous navigation. This last practical guided session consists of implementing basic autonomous navigation tasks (chapter 6b and 6c). Building on the motion module built on previous sessions, the students will implement necessary algorithms for autonomous robots: path planning given a map of the environment and reactive behaviours to avoid unknown obstacles.

Group project - Student robotic contest. Each group will develop a project based on their designed robot and implemented modules during the guided sessions. First they need to integrate all previous modules, and then augment the system with new algorithms proposed by each group to achieve a series of tasks to be performed in a common scenario. Besides, the students are provided with a

Table 2. Details of student working hours distribution in the course

Activity	Hours
Classroom sessions (including discussions and presentations)	30
Guided practical sessions in the laboratory	12
Attendance to seminars from guest robotics researchers	4
Autonomous work in the laboratory or office hours (including discussions with the supervisor)	15/20
Work out of the laboratory (preparation of reports, exercises, studying)	15/20

Table 3. Details of the grading criteria at the different student evaluation activities

Activity	Grade	Description - criteria
Practical sessions	30%	Demonstration and oral discussion with the instructor about the modules developed in practical sessions P2, P3 and P4. Besides, student should hand in reports of preparation work for the practical session and brief post-session report explaining the implemented module.
Group Project	30%	Demonstration of the team project developed functionalities, i.e., demonstrate how many tasks from the contest the robot is able to achieve (each team individually, still out of the contest execution).
Competition results	10%	Contest results achieved by each team as explained in Section 4.
Classroom discussion	10%	Oral presentation and discussions in the classroom sessions, including a final presentation of the team contest achievements.
Classroom exercises	10%	Exercises and problems solved from the proposed exercises in the classroom sessions. These are related to demonstrate the understanding of the concepts explained in the classes or from research papers proposed to be read.
Robot design	10%	Originality and/or quality of the robot designed by each team.
Optional tasks	10%	This additional part can be obtained in the final grade by implementing additional and innovative tasks in the project team, different from the required ones (for example integrating different available sensors, such as a gyro, or implementing new modules for additional skills to demonstrate in the given scenario or out of it).

scenario where to implement and test optional tasks from the more advanced concepts at the course (chapter 7). The details of this contest (scenario, tasks,...), which is being currently developed by our students, are given in the following section 4. The performance of each team will be evaluated individually and in a contest among all the teams, as an additional motivation for innovative and efficient solutions.

3.4 Course Evaluation

The course is designed with the idea that each student should approximately invest the hours detailed in Table 2 in each activity, and each of these activities will be evaluated along the quarter.

Students are required to hand in some exercises before each laboratory session, and perform a demonstration or report after it. Besides, each student will be evaluated during the classroom sessions, through participation in discussions and exercises proposed and through the oral presentations of the modules developed in the practical sessions. Finally the group project contest will be the main evaluation activity, where all modules developed in the practical sessions and project should be demonstrated. The students should prove their project functionality in the contest and also make a presentation describing the interesting or key ingredients in their design. A more detailed distribution of the grading evaluation is shown next in Table 3. Due to the sometimes random component in the contest results, we provide the students with additional ways to obtain the maximum grade with optional activities. This helps to avoid the fact that only the winners of the race achieve the maximum final grades.

4 Results of the Course and Student Robotic Contest

This section describes the results after deploying this course during the last three academic years, paying special attention to the final robotic contest run among the students at the end of the course.

4.1 Results from the Course

Results after running this course for the last three years are very satisfactory: students get very involved in the course, they obtain good results and grades and their interest on robotics activities is also increased after the course, with many of them performing following courses and master thesis in related topics. We have around 15 or 20 students in a class, working in teams of two or three students. Average grade obtained in two previous years (current year is still to be evaluated) is 8 (10 is the maximum grade). Most of the 15 teams that had taken part in the course during previous years were able to execute most of the tasks in the final competition.

4.2 Robotic Competition: Scenario, Tasks and Results

The basic idea of the competition is to integrate all the modules developed in the practical sessions. Then, students should program some additional skills for each robot to take part in the contest-race among all teams: robots should perform as fast as possible, and as accurate as possible, all the required tasks. Each year, we adapt or slightly change the required tasks to be performed during the competition but the basic ideas are similar. The scenario built for last year competition and a detailed scheme of its different areas are shown in Fig. 3.

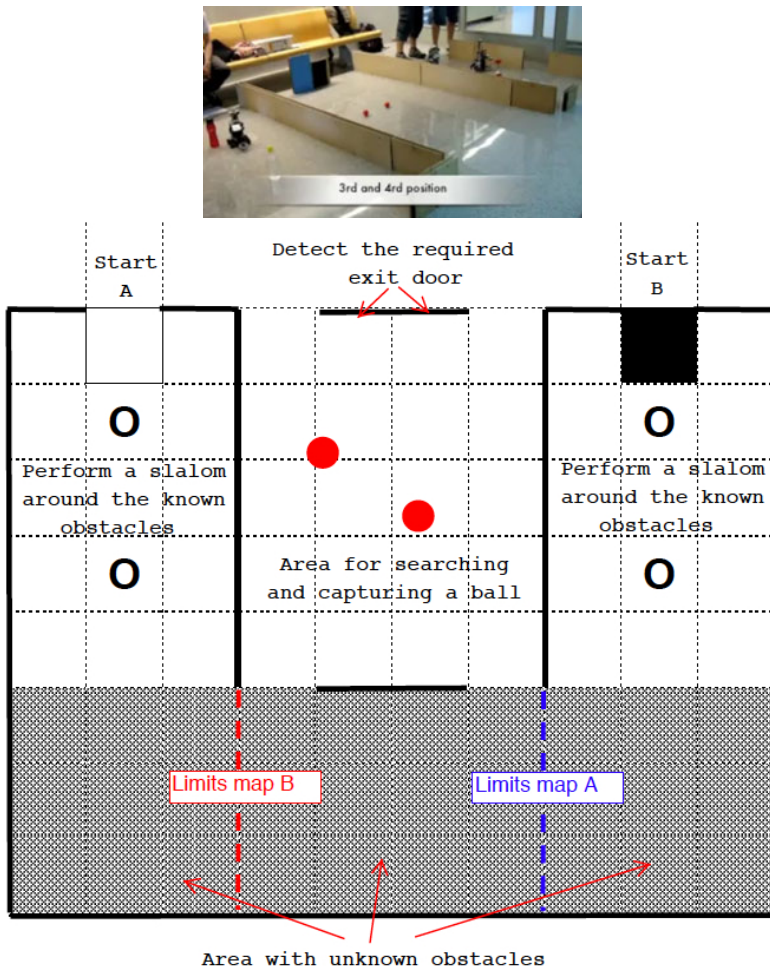


Fig. 3. An image of the real scenario built during the competition (top) and a scheme of the contest scenario (bottom), pointing in which parts of the scenario the robot should perform each of the tasks

Required tasks. Given a map of the environment, each team has to be able to perform the following tasks in specific areas of the scenario:

- Detect the starting location from the ground color. This would allow the robot to initialize its starting location coordinates, its goal location coordinates and the goal-color.
- Perform a "slalom" through known obstacles in part A of the scenario.
- Traverse the maze in part B of the scenario. The map only provides the out-limits, but unknown obstacles can appear, therefore the robot should be able to avoid them and re-plan its way out of the maze.

- Find and catch a ball, using the camera and some kind of clamp or grasping mechanism, in part C of the scenario.
- While carrying the ball, the robot should identify using the camera which is the exit-door of its given color and get out through it.

Designed robots. Students are told to build a robot with these requirements:

- The motion should be based on wheels, fitting approximately the differential drive kinematics model.
- The third motor should be used to close a grasping mechanism, and the camera should be placed pointing towards it.
- The robot should include as well a reflected light sensor pointing to the floor to detect signs in the ground, and a sonar sensor pointing to the front to avoid obstacles.

Figure 4 shows some examples of the different robot constructions built by the students of this course. Most of these robots were able to perform all tasks. Initially, every robot has to demonstrate individually all the activities that it is able to perform properly. Then, sorting the participants according to the time needed to perform all the tasks, all the teams take part in a kind of play-off: pairs of robots compete against each other, and the winner of each round qualifies for the next. Figure 5 shows two robots searching and capturing the ball, in the scenario built for the competition. A more detailed summary of the achievements can be observed in the videos available in the course website¹. The competition evaluation considers not only the fastest participant on getting out of the scenario, but also how good the tasks were achieved on the way.

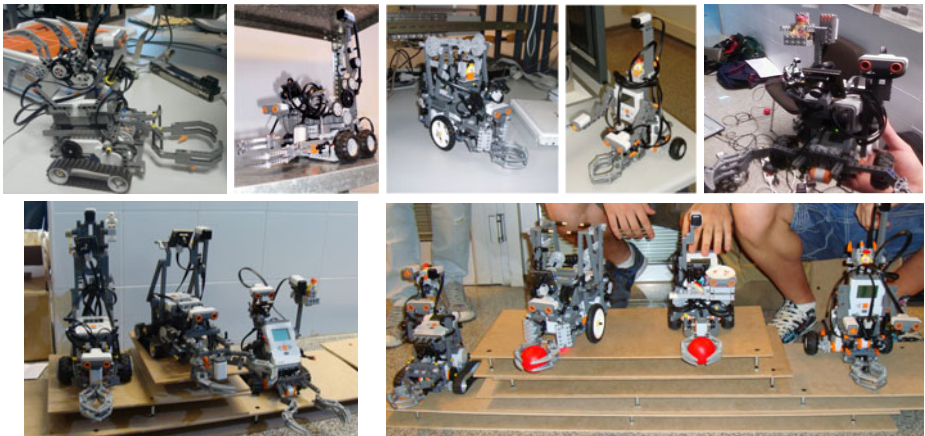


Fig. 4. Examples of the robots built by the students and podiums of two previous years with winners of the race

¹ <http://webdiis.unizar.es/~anacris/carrera10/index.html>

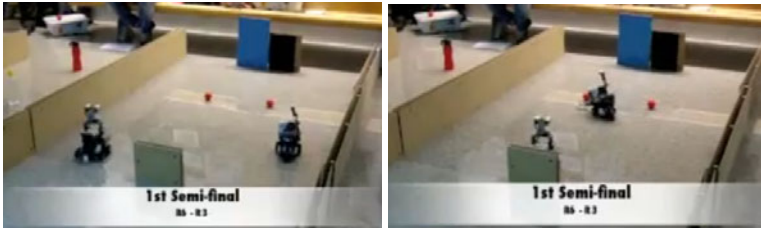


Fig. 5. Video frames of a contest round with two participants in the scenario during searching phase (left) and trying to catch the ball (right)

5 Advanced Robotics Experimentation

For students interested in robotic, we offer them to perform projects/thesis on advanced robotics topics and solve more complicated tasks. The growing popularity of the Mindstorms platform makes possible the use of research- and industry-grade tools as a transitional step, by using the same software suites that are deployed in more powerful robots. Two approaches are possible: on the one hand, the use of regular software in small laptops on-board the robot or nearly located using Bluetooth; on the other, the replacement of the stock NXT firmware with an embedded controller. From the most extended ones in the robotics community, two of each of these options are available to enthusiast alumni:

1. *Player robotic library* [4]. This library provides an abstract layer (via so-called interfaces) over a large variety of robotic platforms. It is widely used in the research community for both simulation and real experimentation. A driver (nxt) developed by the authors and available right now in the project SVN enables control of the NXT actuators via direct USB connection.

2. *ROS – Robot Operating System* [11]. This newer robotic framework is well-suited for complex robots, with many degrees of freedom, and possibly using a cluster of embedded machines instead of a single computer for control. Software modules in ROS are structured in *stacks*, such as NXT-ROS which enables full interaction with a NXT brick.

3. *Lejos – Java for LEGO Mindstorms*.² Moving into the embedded domain, this project delivers a tiny Java virtual machine that replaces the stock firmware and a suite of accompanying tools. Thus, it enables the use of Java and its modern characteristics on-board the robot. A class library is also provided that gives access to the NXT sensors and actuators.

4. *GNAT GPL Ravenscar edition for Mindstorms*.³ Advanced topics arising in high-reliability and safety embedded systems can be approached by means of the software suite made available under GPL license by AdaCore. This tool, which also replaces the stock NXT firmware, enables usage of Ada, a language

² Available at <http://lejos.sourceforge.net>

³ Available at <http://libre.adacore.com/libre/tools/mindstorms/>

well-suited for high-integrity systems; SPARK, an Ada subset aimed at automatic code validation; and the Ravenscar profile, designed for safety-critical real-time software that has to undergo certification.

As initial results regarding these advanced topics, the following activities have been finished:

- One student designed during his Master thesis an environment to remotely monitor and control Mindstorms robots with LeJOS, including algorithms from the course and additional motion models and algorithms regarding planning and avoidance.
- The authors working in multi-robot cooperation have prepared a LEGO model which houses a regular nettop and a small Hokuyo laser rangefinder. With this setup, localization and communication can be easily achieved in indoor setups, offering the possibility of Master-level research projects using the mentioned tools, under the guidance and expertise of the researchers in the department.

6 Conclusions

This paper summarizes framework, goals and contents of our service robotics course. This course is offered in our engineering degrees, since mobile robotics and autonomous systems are subjects of increasing interest in the curricula of engineering studies, being application domains of growing importance in leading research and industry trends. The course is given in a multi-disciplinary class environment and has a strong practical component. The practical part of the course is based on LEGO Mindstorms, which the students use to freely build a robot to perform the learned and required tasks and take part in the final course contest. This strong practical component has been shown in our experience to be a highly motivating factor for the students, specially the final course contest. The results have been really interesting, with very good final gradings both for the students (average of 8.5 out of 10) and the course (the students highly value this course).

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References

1. Benedettelli, D., Casini, M., Garulli, A., Giannitrapani, A., Vicino, A.: A LEGO Mindstorms experimental setup for multi-agent systems. In: 3rd IEEE Multi-Conference on Systems and Control, pp. 1230–1235 (2009)
2. Druin, A., Hendler, J. (eds.): Robots for kids: exploring new technologies for learning. Morgan Kaufmann Publishers Inc., San Francisco (2000)

3. Gawthrop, P., McGookin, E.: Using LEGO in control education. In: Dormido, S., Morilla, F., Sanchez, J. (eds.) 7th IFAC Symposium on Advances in Control Education. IFAC, Madrid (2006)
4. Gerkey, B.P., Vaughan, R.T., Howard, A.: The player/stage project: Tools for multi-robot and distributed sensor systems. In: Int. Conf. on Advanced Robotics, pp. 317–323 (2003)
5. Goncales, J., Lima, J., Malheiros, P., Costa, P.: Realistic simulation of a LEGO Mindstorms NXT based robot. In: Control Applications (CCA) Intelligent Control (ISIC), pp. 1242–1247. IEEE, Los Alamitos (2009)
6. Grega, W., Pilat, A.: Real-time control teaching using LEGO® Mindstorms®. In: IMCSIT, pp. 625–628 (2008)
7. Kim, S.H., Jeon, J.W.: Educating C language using LEGO Mindstorms robotic invention system 2.0. In: Proc. of IEEE International Conference on Robotics and Automation, pp. 715–720 (2006)
8. Klassner, F., Continanza, C.: Mindstorms without robotics: an alternative to simulations in systems courses. SIGCSE Bull. 39, 175–179 (2007)
9. Leitão, P., Gonçalves, J., Barbosa, J.: Learning mobile robotics using LEGO Mindstorms. In: 9th Spanish-Portuguese Congress on Electrical Engineering (2005)
10. McWhorter, W.I., O'Connor, B.C.: Do LEGO® Mindstorms® motivate students in CS1? SIGCSE Bull. 41, 438–442 (2009), <http://doi.acm.org/10.1145/1539024.1509019>
11. Quigley, M., Gerkey, B., Conley, K., Faust, J., Foote, T., Leibs, J., Berger, E., Wheeler, R., Ng, A.: ROS: an open-source robot operating system. In: IEEE/RSJ Int. Conf. on Robotics and Automation (ICRA 2009): Workshop on Open Source Software (2009)
12. RobotC, <http://www.robotc.net>, carnegie Mellon University - Robotics Academy
13. Talaga, P., Oh, J.C.: Combining AIMA and LEGO Mindstorms in an artificial intelligence course to build real world robots. J. Comput. Small Coll. 24, 56–64 (2009)